

Week 0: Course Overview & Strategic Diagnosis

DOT E 3090: Game Theory for Business Strategy

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Make better decisions when other decision-makers are thinking too.

Your First Strategic Decision

The Situation

You and a competing firm choose prices **simultaneously**. Neither firm observes the other's choice before deciding.

	Rival: Maintain	Rival: Cut
You: Maintain	(8, 8)	(2, 12)
You: Cut	(12, 2)	(4, 4)

What would you choose? What reasoning supports your decision?

Payoffs represent profits; your payoff is listed first.

You Have Just Used Game Theory

- There were identifiable **players**.
- Each player had available **strategies**.
- Each outcome produced a pair of **payoffs**.
- The value of your choice depended on the rival's choice.
- Individually sensible choices might produce an outcome both firms dislike.

Strategic Interdependence

Your outcome depends on your decision **and** on decisions made by others.

This is the central problem that game theory helps us analyse.

What Game Theory Is and Is Not

It Is Not

- a collection of tricks for defeating rivals;
- a guarantee that one best move always exists;
- mathematics disconnected from business decisions.

It Is

A disciplined way to reason when outcomes depend on interacting decisions.

- Anticipate strategic responses.
- Identify stable outcomes and strategic traps.
- Explore how changing incentives or choices changes behaviour.

The Business Questions We Will Investigate

- Why do firms cut prices even when a price war hurts both sides?
- When can a major investment deter entry, and when can it backfire?
- How should a company bid in an auction without overpaying?
- What determines bargaining power in an acquisition, supplier contract, or venture-capital deal?
- How do platforms manage competition and cooperation among consumers, suppliers, and business partners?

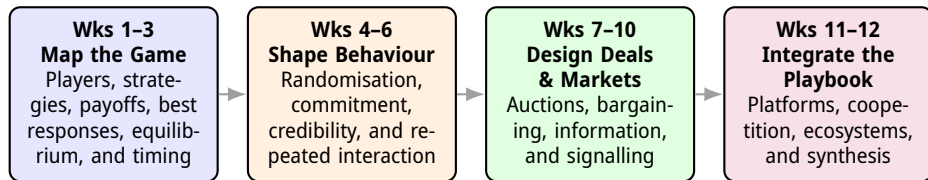
What Will You Be Able to Do?

- 1 **Diagnose** a business situation by identifying the relevant players, choices, incentives, information, and timing.
- 2 **Analyse** likely behaviour and outcomes using appropriate game-theoretic tools.
- 3 **Evaluate** how competition, cooperation, negotiation, and strategic communication shape business outcomes.
- 4 **Defend** a practical recommendation by stating its logic, assumptions, and likely responses.

The Central Skill

Make better decisions when your result depends on what other decision-makers do.

The Course Playbook



Week 13: Final Examination

Apply the course tools to open-book, case-based strategic problems.

How Each Class Works

1. Understand

Learn the concept, model, and analytical tool through intuitive explanation and formal reasoning.

2. Examine

Investigate how the strategic logic appears in a contemporary business case.

3. Apply

Test your reasoning through cases, simulations, negotiations, worksheets, and discussion.

Understand the model. See it in business. Use it yourself.

The Strategic Diagnosis We Will Reuse All Term

- 1 **Decision:** Who must decide what?
- 2 **Game:** Who are the players, and what can each choose?
- 3 **Structure:** What are the timing, information, and incentives?
- 4 **Analysis:** What responses and outcomes does the model predict?
- 5 **Recommendation:** What should the decision-maker do, and what could reverse the conclusion?

Modelling Discipline

Select the model only after diagnosing the strategic problem.

How Your Learning Is Assessed

Component	What It Assesses
Examination (40%)	Open-book, case-based examination assessing understanding and application of game-theory concepts and analytical tools. <i>(LO1, LO3)</i>
Project (20%)	Team-based analysis and presentation of a recent business news event, focusing on strategic interaction, incentives, responses, and outcomes. <i>(LO2, LO3)</i>
Continuous Assessment (40%)	Weekly in-class cases, strategic mapping exercises, simulations, negotiations, discussions, and other learning activities. <i>(LO1, LO2, LO3)</i>

What Strong Work Demonstrates

Appropriate model choice, explicit assumptions, accurate analysis of likely responses, and a practical recommendation supported by the model.

Use of AI Tools and Academic Integrity

Course Policy: Approach 1

AI tools may not be used in any graded learning activity or assessment in this course.

- Submit work based on your own independent analysis.
- This prohibition includes generating, rewriting, summarising, or analysing assessed work.
- If you are uncertain whether a tool or feature is permitted, ask **before** using it.
- Unauthorised use may constitute academic dishonesty under University procedures.

Consult the official course outline and University guidance for the complete policy, declaration, submission, and academic-honesty requirements.

Before Week 1

- 1 Complete the assigned Week 1 pre-reading on the course site.
- 2 Bring one example of a business decision whose result depends on another decision-maker's response.
- 3 For your example, identify the **players**, their possible **strategies**, and what each player values.

Next Week

We return to the pricing game and learn how to solve it systematically.

Your decision changes the incentives of others.

Their response may change your best decision.

Think strategically, not in isolation.

Make better decisions when other decision-makers are thinking too.

Welcome to DOTE 3090.

Supplementary: Course at a Glance

Week	Strategic Tool	Illustrative Applications
1	Strategic interaction	Pricing, promotion, and rivalry
2	Sequential moves	Entry, timing, and strategic response
3	Simultaneous moves	Nash equilibrium and competitive choices
4	Mixed strategies	Promotions, advertising, and inspection
5	Commitment	Capacity, infrastructure, and content
6	Repeated games	Ongoing rivalry and cooperation
7–8	Auctions	Land, spectrum, procurement, and platforms
9	Bargaining	M&A, suppliers, and venture capital
10	Information and signalling	Warranties, IPOs, credit, and collateral
11	Platforms and coopetition	Delivery, marketplaces, and regulation
12	Strategic synthesis	Ecosystems and multi-player strategy
13	Final examination	Open-book, case-based application

Supplementary: Two Tools You Will Build

Payoff Matrix

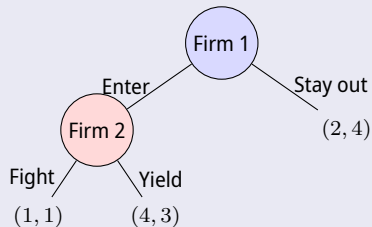
Represents outcomes when players choose without observing one another's current choices.

	Maintain	Cut
Maintain	(8, 8)	(2, 12)
Cut	(12, 2)	(4, 4)

Introduced formally in Week 1.

Game Tree

Represents the order of decisions and how later players can respond.



Introduced formally in Week 2.

The representation should match the timing and information structure of the decision.

Supplementary: How to Study with These Slides

- 1 Answer the visible question before reading the notes.
- 2 Use the notes to check definitions, assumptions, and reasoning.
- 3 Reconstruct the matrix, tree, calculation, or strategic argument without looking.
- 4 Change one assumption and explain whether the prediction changes.
- 5 Finish with a one-sentence recommendation and one limitation.

Mastery Test

You understand a tool when you can explain why it fits, use it correctly, predict a strategic response, and state when the recommendation could fail.